

File change semantics: context and composition

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“Consider a device designed to read a text in some natural language, interpret it, and store the content in some manner, say, for the purpose of being able to answer questions about it. To accomplish this task, the machine will have to fulfill at least the following basic requirement. It has to be able to build a file that consists of records of all the individuals, that is, events, objects, etc., mentioned in the text and, for each individual, record whatever is said about it. Of course, for the time being at least, it seems that such a text interpreter is not a practical idea, but this should not discourage us from studying in abstract what kind of capabilities the machine would have to possess, provided that our study provides us with some insight into natural language in general.” (Karttunen 1976)

The notion of a *discourse referent* emerged from the work of Lauri Karttunen and David Lewis (Karttunen 1976, Lewis 1979), during a time of general optimism concerning connections between linguistic theory and artificial intelligence research.

The central idea is that discourse participants introduce and manipulate *variables* corresponding to individuals mentioned over the course of a conversation.

- (1) A^x woman was bitten by a y dog.
- (2) She_x hit it_y .
- (3) It_y jumped over a z fence.

This powerful idea subsequently informed dynamic approaches to meaning, which essentially model anaphora as a kind of cross-referencing device (Heim 1982, Groenendijk & Stokhof 1991).

In this course, we'll primarily be interested in *dynamic semantics* proper, rather than discourse representation theories (Kamp 1981), which are often grouped under the same umbrella.¹

The dynamic slogan:

“You know the meaning of a sentence if you know the change it brings about in the information state of anyone who accepts the news conveyed by it” (Veltman 1996).

¹See (Chierchia 1995) for a detailed comparison.